

MAT 322 HW02

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Problem 1. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and $g: \mathbb{R}^n \rightarrow \mathbb{R}$ be functions. If f and g are differentiable (respectively, class C^1 on a neighborhood of x) at the point $x \in \mathbb{R}^n$, prove that their sum $f + g$ and product fg are both differentiable (respectively, class C^1 on a neighborhood of x) at x and find their derivatives at x .

Solution. If f and g are differentiable at x , then

$$\begin{aligned}\Delta_f(x, h) &:= f(x + h) - f(x) = (df)_x(h) + \|h\|\varepsilon_f(h) \\ \Delta_g(x, h) &:= g(x + h) - g(x) = (dg)_x(h) + \|h\|\varepsilon_g(h)\end{aligned}$$

for functions $\varepsilon_f, \varepsilon_g: \mathbb{R}^n \rightarrow \mathbb{R}$ whose limits as h approaches zero are both 0. Then, we can see that the sum $f + g$ is differentiable at x with derivative $(d(f + g))_x = \boxed{(df)_x + (dg)_x}$ because

$$(f + g)(x + h) - (f + g)(x) = (df)_x(h) + (dg)_x(h) + \|h\|(\varepsilon_f(h) + \varepsilon_g(h))$$

and $\varepsilon_f + \varepsilon_g$ is zero in the limit as h goes to zero. Similarly, the product fg is differentiable with derivative $(d(fg))_x = \boxed{f(x) \cdot (dg)_x + g(x) \cdot (df)_x}$. To show this, we write

$$\begin{aligned}(fg)(x + h) - (fg)(x) &= (fg)(x + h) - f(x + h)g(x) + f(x + h)g(x) - (fg)(x) \\ &= f(x + h)\Delta_g(x, h) + g(x)\Delta_f(x, h)\end{aligned}$$

Now, we handle these two addends separately. For the first addend, we can expand $f(x + h) = f(x) + (df)_x(h) + \|h\|\varepsilon_f(h)$ and $\Delta_g(x, h) = (dg)_x(h) + \|h\|\varepsilon_g(h)$. This leaves four important pieces:

- $f(x) \cdot (dg)_x(h)$;
- $(df)_x(h) \cdot (dg)_x(h)$;
- $\|h\|\varepsilon_f(h) \cdot (dg)_x(h)$; and
- $\|h\|\varepsilon_g(h) \cdot f(x + h)$.

Of these, I claim that the latter three each decay faster than h as h goes to zero, i.e. they are part of the error term when computing the derivative of fg . We verify this:

- Noting that $(df)_x$ and $(dg)_x$ are linear maps, they are continuous on all of $\mathbb{R}^n$. To verify that

$$\lim_{h \rightarrow 0} \frac{(df)_x(h)}{\|h\|} \cdot (dg)_x(h) = 0,$$

we need only verify that

$$\frac{(df)_x(h)}{\|h\|} = (df)_x \left(\frac{h}{\|h\|} \right)$$

is bounded on a neighborhood of 0 because $(dg)_x(0) = 0$, and this follows from the continuity of linear functions.

- The other two terms are easier: they already come with a $\|h\|$ attached, so we only need to check that

$$\lim_{h \rightarrow 0} \varepsilon_f(h) \cdot (dg)_x(h) = \lim_{h \rightarrow 0} \varepsilon_g(h) \cdot f(x+h) = 0.$$

Since $\varepsilon_f(h)$ and $\varepsilon_g(h)$ already approach 0 with h , it suffices to check that $(dg)_x(h)$ and $f(x+h)$ are bounded on a neighborhood of 0, but again this follows from continuity of the functions $(dg)_x$ and f (the latter is continuous at x because it is differentiable at x).

We will denote the sum of these three terms with $\|h\|\varepsilon_1(h)$, keeping in mind that

$$\lim_{h \rightarrow 0} \varepsilon_1(h) = 0$$

The other addend is much simpler to work with: it expands as

$$g(x) \cdot (df)_x(h) + \|h\|\varepsilon_f(h) \cdot g(x).$$

Noting that $g(x)$ is constant with respect to h , we directly have

$$\lim_{h \rightarrow 0} \varepsilon_f(h) \cdot g(x) = g(x) \lim_{h \rightarrow 0} \varepsilon_f(h) = 0.$$

We will write $\varepsilon_f(h)g(x) = \varepsilon_2(h)$, noting also that $\varepsilon_2(h)$ approaches 0 as h approaches 0.

Thus, we can conclude that

$$(fg)(x+h) - (fg)(x) = f(x) \cdot (dg)_x(h) + g(x) \cdot (df)_x(h) + \|h\|(\varepsilon_1(h) + \varepsilon_2(h)),$$

which means that, by the definition of derivative, the derivative of fg at x is $f(x) \cdot (dg)_x + g(x) \cdot (df)_x$.

In the case that f and g are both class C^1 on a neighborhood of x , we can use the fact that class C^1 functions are differentiable on this neighborhood together with the new derivative formulas we have just proven above to show that $f+g$ and fg are also class C^1 . Specifically, the Jacobian matrices of $(d(f+g))_x$ and $(d(fg))_x$ precisely contain the partial derivatives of their respective functions, so we just need to verify that these entries are all continuous; but this is obvious for sums because

$$(d(f+g))_x = (df)_x + (dg)_x$$

and so the relevant partials (matrix entries) are sums of continuous functions and it is obvious for products because

$$(d(fg))_x = f(x) \cdot (dg)_x + g(x) \cdot (df)_x,$$

where the partials look like

$$f(x) \cdot \frac{\partial g}{\partial x_j} + g(x) \cdot \frac{\partial f}{\partial x_j},$$

which are continuous because they are sums of products of continuous functions. $\square$

Problem 2. Show that the function $f(x, y) = |xy|$ is differentiable at $(0, 0)$, but it is not of class C^1 on any open neighborhood of $(0, 0)$.

Solution. I claim that $(df)_{(0,0)}$ is the zero map. To check this we need to show that

$$\lim_{(\alpha, \beta) \rightarrow (0,0)} \frac{f(\alpha, \beta) - f(0, 0) - (df)_{(0,0)}(\alpha, \beta)}{\|(\alpha, \beta)\|} = \lim_{(\alpha, \beta) \rightarrow (0,0)} \frac{|\alpha\beta|}{\sqrt{\alpha^2 + \beta^2}} = 0.$$

By definition, this means we want to show that for each $\varepsilon > 0$, there exists a $\delta > 0$ such that $\|(\alpha, \beta) - (0, 0)\| = \sqrt{\alpha^2 + \beta^2} < \delta$ implies $\frac{|\alpha\beta|}{\sqrt{\alpha^2 + \beta^2}} < \varepsilon$. By the AM-GM inequality,

$$\alpha^2 + \beta^2 \geq 2|\alpha\beta|$$

for any reals α, β , so, taking $\delta = 2\varepsilon$, we get

$$\sqrt{\alpha^2 + \beta^2} < \delta \implies \frac{|\alpha\beta|}{\sqrt{\alpha^2 + \beta^2}} \leq \frac{\frac{1}{2}(\alpha^2 + \beta^2)}{\sqrt{\alpha^2 + \beta^2}} = \frac{\sqrt{\alpha^2 + \beta^2}}{2} < \frac{1}{2}\delta = \varepsilon.$$

Therefore, the limit is 0, so f is differentiable at $(0, 0)$.

However, f does not have continuous partials on a neighborhood of $(0, 0)$, since, for example, $\frac{\partial f}{\partial x}(0, y)$ doesn't even exist for $y \neq 0$. $\square$

Problem 3. Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by setting $f(0) = 0$ and

$$f(t) = t^2 \sin(1/t) \text{ if } t \neq 0.$$

- Show that f is differentiable at 0, and calculate $f'(0)$.
- Calculate $f'(t)$ if $t \neq 0$.
- Show f' is not continuous at 0.
- Conclude that that f is differentiable but not continuously differentiable on $\mathbb{R}$.

Solution. The derivative of f at 0 is

$$\lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t - 0} = \lim_{t \rightarrow 0} t \sin(1/t) = 0,$$

where the second equality uses the fact that $\sin(1/t)$ is bounded on a neighborhood of 0, since $|\sin(x)| \leq 1$ for all $x \in \mathbb{R}$. When $t \neq 0$, the derivative can be computed with the standard derivative laws to get

$$f'(t) = t^2 \cos(1/t) \cdot \frac{-1}{t^2} + 2t \sin(1/t) = 2t \sin(1/t) - \cos(1/t).$$

As t approaches 0, we have

$$\lim_{t \rightarrow 0} f'(t) = \lim_{t \rightarrow 0} 2t \sin(1/t) - \cos(1/t).$$

When t is small, $2t \sin(1/t)$ will become arbitrarily small (it is bounded in absolute value by $|2t|$). Thus, the limit equals

$$-\lim_{t \rightarrow 0} \cos(1/t) = -\lim_{y \rightarrow \infty} \cos(y),$$

but this doesn't exist, so f' cannot be continuous at 0. Therefore, f is differentiable, but its derivative isn't continuous, so it is not class C^1 . $\square$

Problem 4. Let $f: U \rightarrow \mathbb{R}^m$ be a function defined on an open set $U \subseteq \mathbb{R}^m$, such that all partial derivatives $\partial f / \partial x_i$ exist and are bounded in a neighborhood of a point $x \in U$. Prove that f is continuous at x .

Solution. To start with, we reduce to the $m = 1$ case via the following claim:

Claim 4.1 — The function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous at a point x if each component function $f_i: \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous at x .

Proof. If $f_1, \dots, f_m: \mathbb{R}^n \rightarrow \mathbb{R}$ are all continuous at x , then the linear combination

$$f = f_1 e_1 + f_2 e_2 + \dots + f_m e_m: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

is continuous at x because it is a sum of continuous functions at x . □

Thus, it suffices to solve the problem for functions $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and then use the claim to patch together m different $\mathbb{R}^n \rightarrow \mathbb{R}$ functions for the general setting. We will show in that

$$\lim_{h \rightarrow 0} f(x + h) - f(x) = 0.$$

Define

$$v_i = h_1 e_1 + h_2 e_2 + \dots + h_i e_i$$

for $i = 1, 2, \dots, n$, so $h = v_n$, and let $v_0 = 0$. Then, write

$$f(x + h) - f(x) = \sum_{i=1}^n f(x + v_i) - f(x + v_{i-1}).$$

Since each of the partials of f exists on a neighborhood N of x , we can use the mean value theorem to find a value of t_i for $i = 1, 2, \dots, n$ such that

$$f(x + v_i) - f(x + v_{i-1}) = \frac{\partial f}{\partial x_i}(x + v_{i-1} + t_i e_i) \cdot h_i$$

where $0 < t_i < h_i$ as long as each v_i lies in N (which just requires $\|x - h\|$ to be sufficiently small). Since all the partials are bounded on this neighborhood, we can find a constant C such that, for each i ,

$$\left| \frac{\partial f}{\partial x_i}(x + v_{i-1} + t_i e_i) \right| < C.$$

Then, we simply have

$$\lim_{h \rightarrow 0} f(x + h) - f(x) = \lim_{h \rightarrow 0} \sum_{i=1}^n f(x + v_i) - f(x + v_{i-1}) < \lim_{h \rightarrow 0} C \cdot (h_1 + \dots + h_n) = 0,$$

hence f is continuous at x if its partials exist and are bounded on a neighborhood of x . □

Problem 5. Suppose $f: U \rightarrow \mathbb{R}^m$ and $g: V \rightarrow \mathbb{R}^\ell$ are class C^1 functions defined on open sets $U \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$ such that $f(U) \subseteq V$. Prove that the composition $g \circ f: U \rightarrow \mathbb{R}^\ell$ is also C^1 .

Solution. Since f and g are class C^1 , they are differentiable, so their composition is also differentiable, and we can use the chain rule to compute that the derivative of $g \circ f$ at a point x is

$$(d(g \circ f))_x = (dg)_{f(x)} \cdot (df)_x.$$

The entries of the Jacobian matrices of $(dg)_{f(x)}$ and $(df)_x$ are all continuous functions in x since f is differentiable-hence-continuous and f, g are class C^1 , so the product of the matrices has entries that are also continuous in x because these entries are sums of products of continuous functions. $\square$

Problem 6. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ and $g: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the functions

$$\begin{aligned} f(x) &= (e^{2x_1+x_2}, 3x_2 - \cos x_1, x_1^2 + x_2 + 2) \\ g(y) &= (3y_1 + 2y_2 + y_3^2, y_1^2 - y_3 + 1). \end{aligned}$$

(a) Compute $(d(g \circ f))_0$ without computing $(g \circ f)(x)$.

(b) Compute $(d(f \circ g))_0$ without computing $(f \circ g)(x)$.

Solution. We have $f(0, 0) = (1, -1, 2)$ and $g(0, 0, 0) = (0, 1)$. In general,

$$(df)_x = \begin{pmatrix} 2e^{2x_1+x_2} & e^{2x_1+x_2} \\ \sin x_1 & 3 \\ 2x_1 & 1 \end{pmatrix}, \quad (dg)_x = \begin{pmatrix} 3 & 2 & 2y_3 \\ 2y_1 & 0 & -1 \end{pmatrix},$$

so we can compute the following four derivatives:

$$\begin{aligned} (df)_0 &= \begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix} \\ (df)_{g(0)} &= \begin{pmatrix} 2e & e \\ 0 & 3 \\ 0 & 1 \end{pmatrix} \\ (dg)_0 &= \begin{pmatrix} 3 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} \\ (dg)_{f(0)} &= \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & -1 \end{pmatrix} \end{aligned}$$

Therefore, the answer to (a) is

$$(d(g \circ f)) = (dg)_{f(0)} \cdot (df)_0 = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} 6 & 13 \\ 4 & 1 \end{pmatrix}}$$

and the answer to (b) is

$$(d(f \circ g)) = (df)_{g(0)} \cdot (dg)_0 = \begin{pmatrix} 2e & e \\ 0 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \boxed{\begin{pmatrix} 6e & 4e & -e \\ 0 & 0 & -3 \\ 0 & 0 & -1 \end{pmatrix}}. \quad \square$$

Problem 7. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at the point $u \in \mathbb{R}^n$, and $b \in \mathbb{R}^m$ a vector. Consider the function $F: \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$F(x) = \|f(x) - b\|^2.$$

Prove that F is differentiable at u and that, for any $v \in \mathbb{R}^n$,

$$(\mathrm{d}F)_u(v) = 2\langle f(u) - b, (\mathrm{d}f)_u(v) \rangle.$$

Solution. Let $g(x) = \|x\|^2 = \langle x, x \rangle$ be a function from $\mathbb{R}^m$ to $\mathbb{R}$.

Claim 7.1 — The function g is differentiable, and the derivative of g at point u is the linear map determined by

$$(\mathrm{d}g)_u(v) = \langle 2u, v \rangle$$

for all $v \in \mathbb{R}^m$.

Solution. This is just a simple computation:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{g(u+h) - g(u) - (\mathrm{d}g)_u(h)}{\|h\|} &= \lim_{h \rightarrow 0} \frac{\langle 2u+h, h \rangle - \langle 2u, h \rangle}{\|h\|} \\ &= \lim_{h \rightarrow 0} \frac{\langle h, h \rangle}{\|h\|} = \lim_{h \rightarrow 0} \frac{\|h\|^2}{\|h\|} = \lim_{h \rightarrow 0} \|h\| = 0. \end{aligned}$$

Note that the matrix for this linear map is $2u^\top$, the transpose of the vector $2u$. $\square$

Now we note that $f(x) - b$ is differentiable at u with derivative $(\mathrm{d}f)_u$ because the constant function sending every point of $\mathbb{R}^n$ to b has a derivative of 0. Therefore, the chain rule says that $F(x) = g(f(x) - b)$, being a composition of functions that are differentiable at u , is also differentiable at u , with derivative

$$(\mathrm{d}F)_u = (\mathrm{d}g)_{f(u)-b} \cdot (\mathrm{d}f)_u.$$

Then, we conclude that

$$(\mathrm{d}F)_u(v) = (\mathrm{d}g)_{f(u)-b} \cdot (\mathrm{d}f)_u(v) = 2(f(u) - b)^\top \cdot (\mathrm{d}f)_u(v) = 2\langle f(u) - b, (\mathrm{d}f)_u(v) \rangle,$$

as desired. $\square$